

SkillWeave: Weaving Heterogeneous Demonstrations into Long-Horizon Manipulation Skills

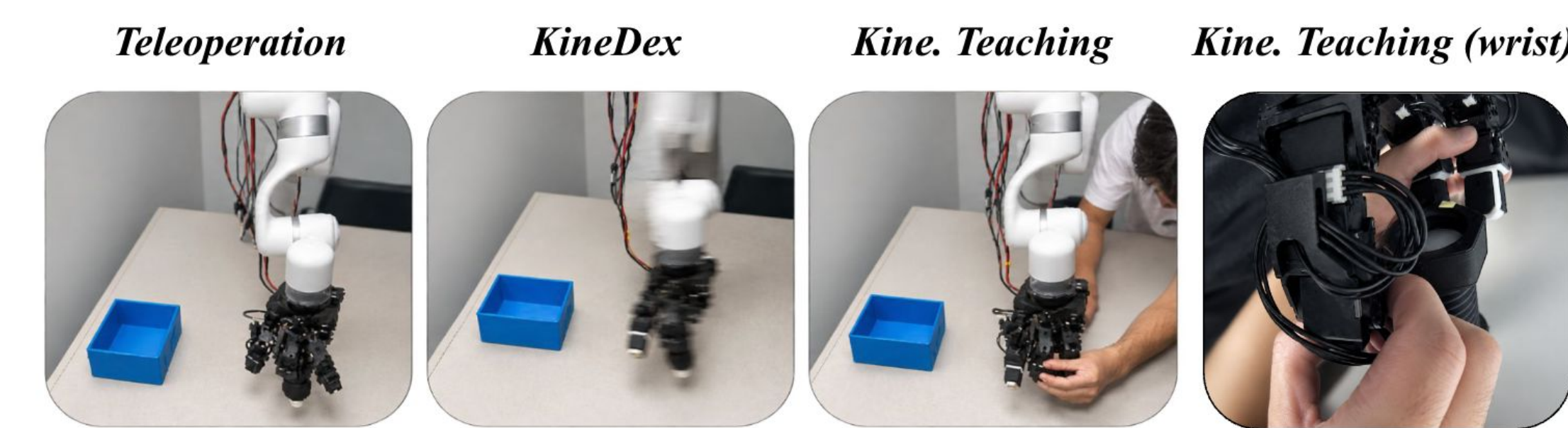
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INTRODUCTION

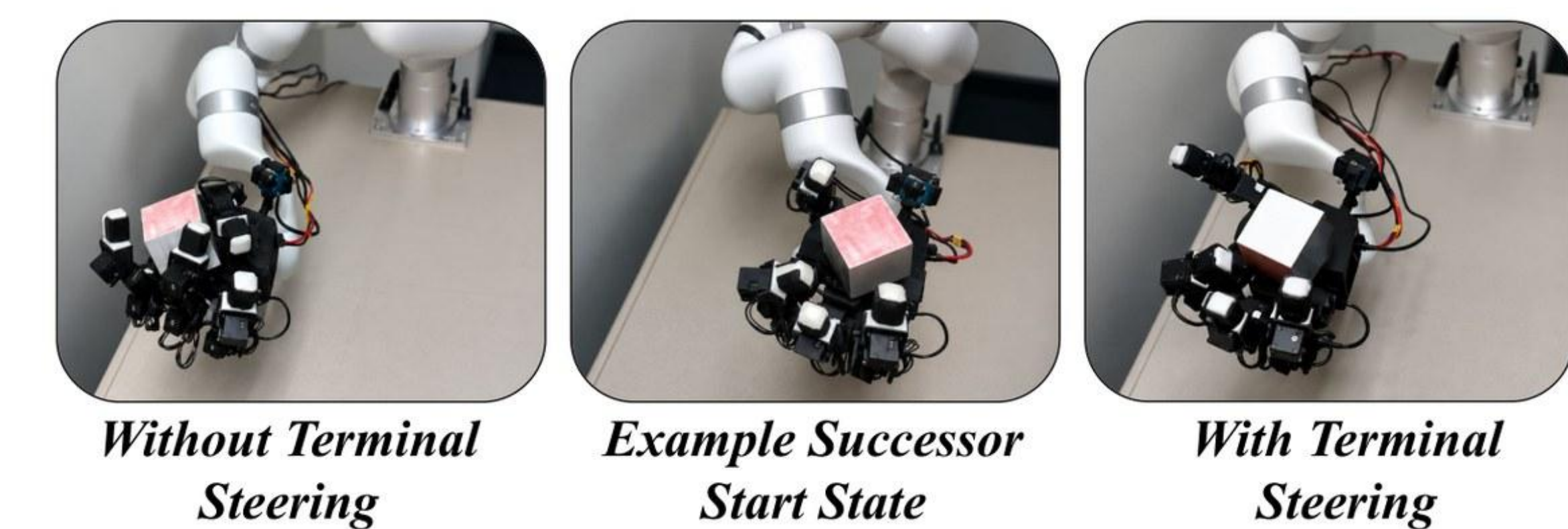
Goal: Long-horizon dexterous manipulation
= **global planning** + **contact-rich precision**

- **Teleoperation:**
 - ✓ Reaching, transport
 - ✗ Contact-rich skills (retargeting error, indirect feedback)
- **Kinesthetic teaching:**
 - ✓ Precise contact
 - ✗ Hands occlude camera, tiring for long demos
- **SkillWeave:**
 - Demonstrate each sub-task in its best-fit modality
 - Train one diffusion policy per sub-task
 - Steer handoffs between policies



CHALLENGES

- **Visual mismatch:**
Demos show the human hand; rollouts never do inpainting (KineDex [1]) → blur, hallucinated content
- **Handoffs:**
Independently trained policies → drift at boundaries
Predecessor end state $\neq$ successor start state



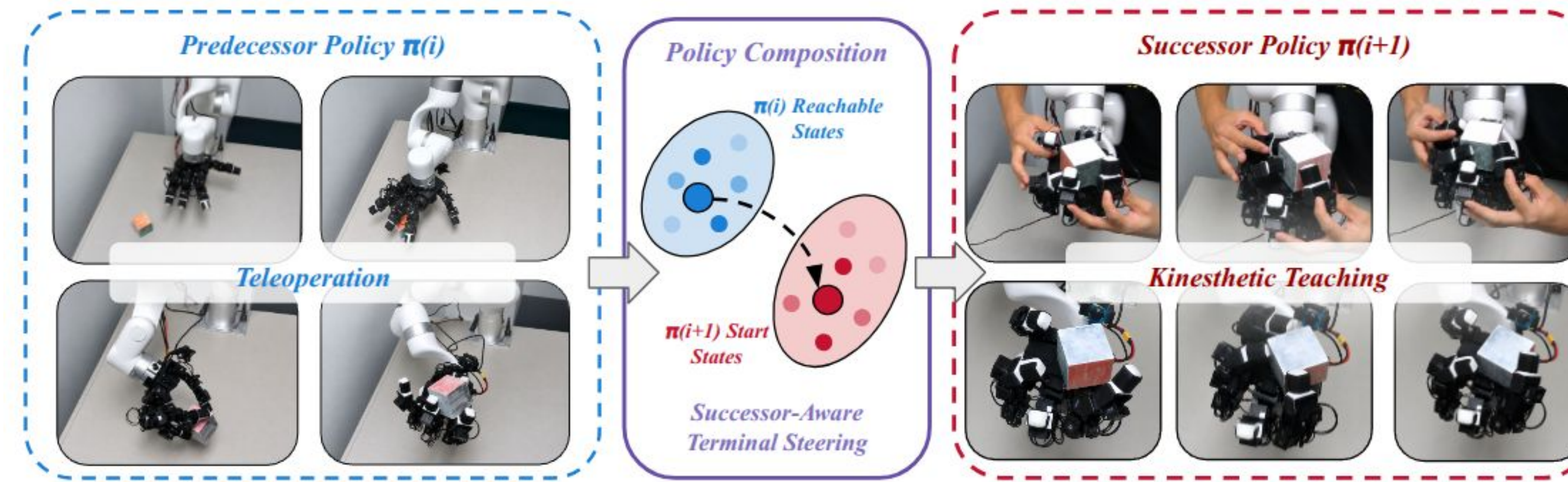
References

- [1] D. Zhang et al. KineDex: tactile-informed visuomotor policies via kinesthetic teaching. CoRL 2025.
[2] C. Chi et al. Diffusion Policy: visuomotor policy learning via action diffusion. RSS 2023.
[3] N. Carion et al. SAM 3: segment anything with concepts. arXiv:2511.16719, 2025.
[4] K. Shaw et al. LEAP Hand: low-cost, efficient, anthropomorphic hand for robot learning. RSS 2023.
[5] C. Lugaresi et al. MediaPipe: a framework for building perception pipelines. arXiv:1906.08172, 2019.

Program participation supported by

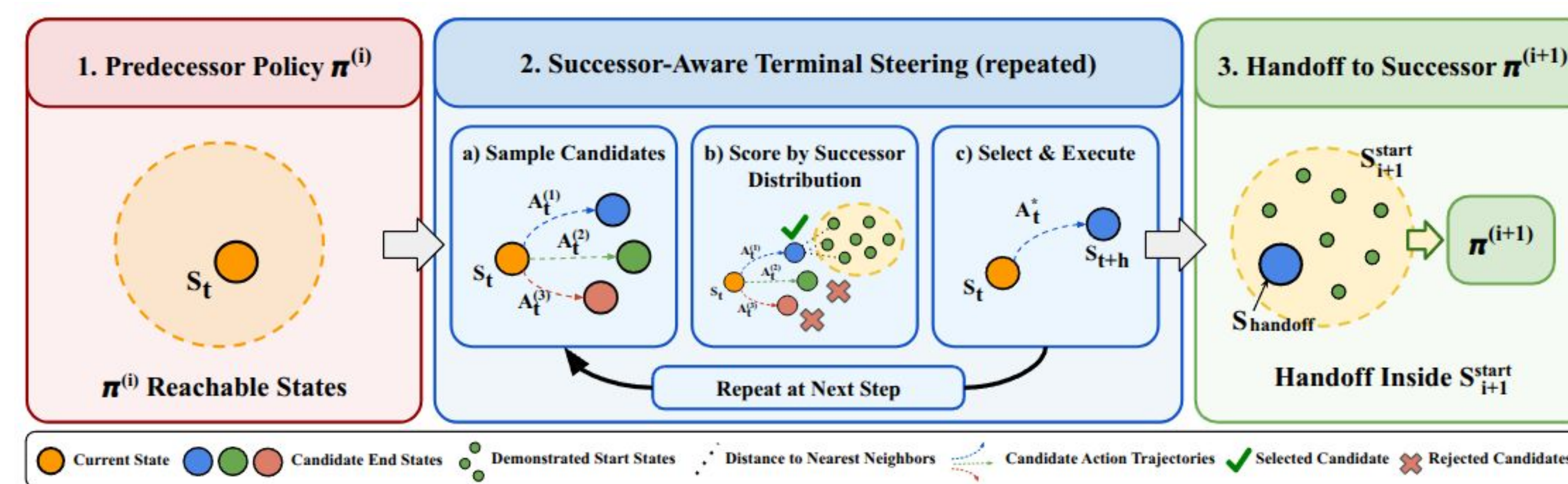
METHODS

Pipeline



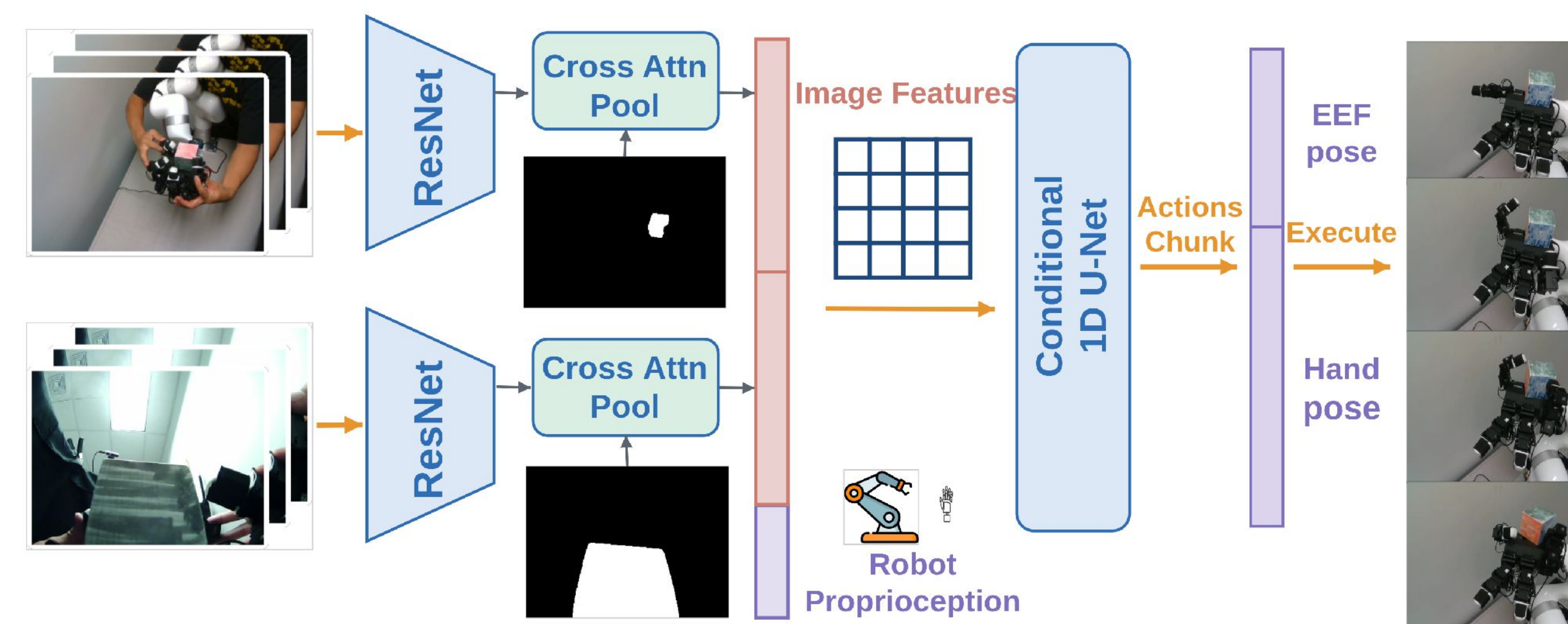
Each task is a chain of sub-tasks, one **diffusion policy** per sub-task. Arm-level sub-tasks use **teleoperation**, contact-rich finger sub-tasks **kinesthetic teaching**; **terminal steering** links the policies.

Terminal steering



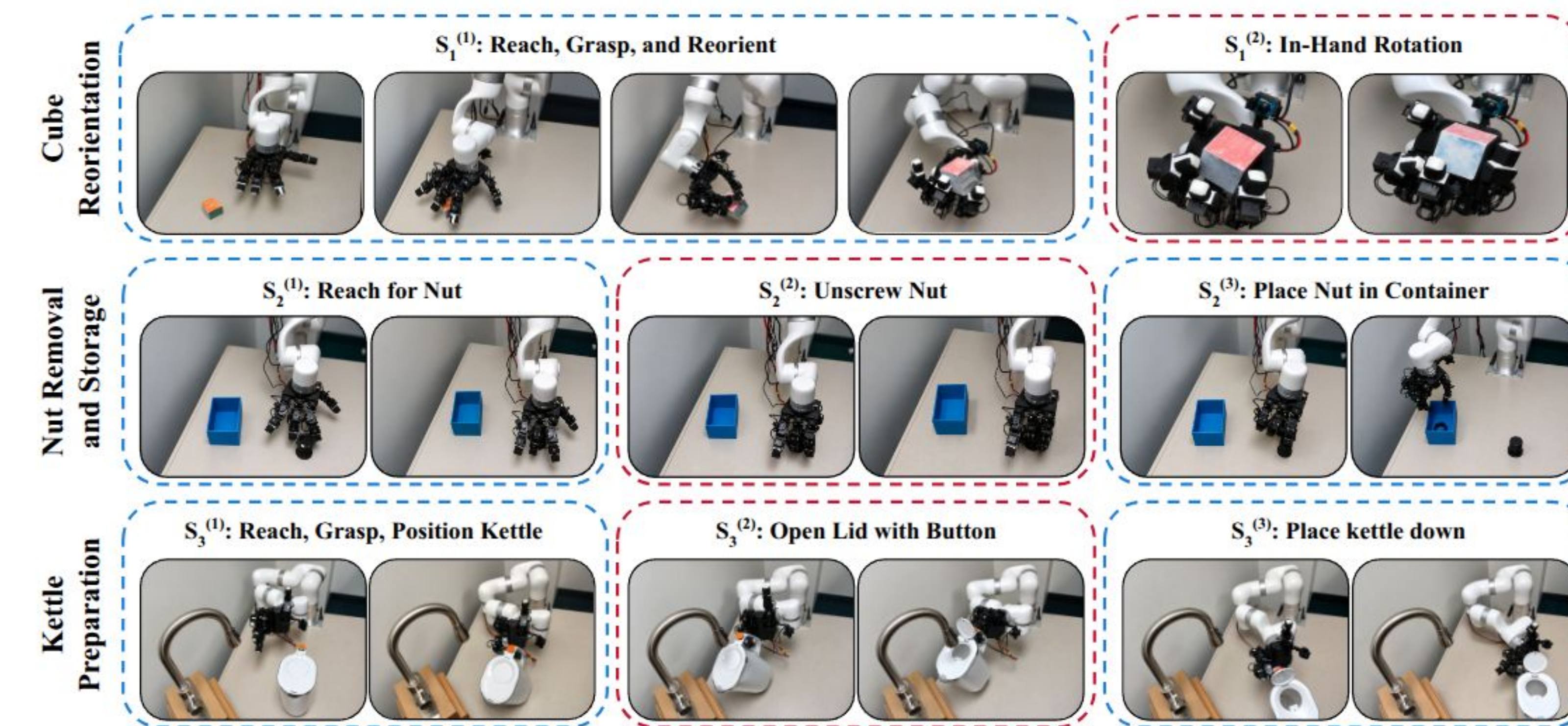
- Samples **L action chunks**, predicts each **endpoint**, scores it by the **distance to the successor's demonstrated start states**.
- Executes the **closest chunk**, repeats each step, hands off once the distance is **below a threshold**.
- **No transition demos, no transition policy:** both policies stay unchanged.

Mask conditioning (kinesthetic sub-tasks)



- Keep the **raw RGB**; features inside the **object mask** form the cross-attention query, and the mask **supervises the attention** in training.
- The policy attends to the **object**, so the demonstrator's hand **does not matter**.
- Masks from **SAM 3** offline for training, a **small U-Net** used for deployment: **no inpainting**, no SAM 3 in the loop.

RESULTS



(a) Sub-task Success

Sub-task	Teleop.	Kin.	KineDex	Mask-Kin.
Reach, grasp, & reorient	70%	0%	0%	—
In-hand rotation	10%	30%	0%	70%
Reach for nut	85%	0%	40%	—
Unscrew nut	5%	0%	60%	75%
Place nut in container	30%	0%	5%	—
Reach, grasp, & position kettle	80%	0%	0%	—
Open lid with button	0%	0%	0%	50%
Place kettle down	55%	0%	0%	—

(b) Long-Horizon Success

Task	Teleop.	Kin.	KineDex	Ours w/o Steering	Ours
Cube Reorientation	0%	0%	0%	10%	45%
Nut Removal & Storage	0%	0%	0%	0%	15%
Kettle Preparation	0%	0%	0%	5%	20%

27%
End-to-end average success
(5% without steering)

65%
Dexterous sub-task average success

87%
Average composition efficiency
(14% without steering)

- **Setup:** 3 real-robot tasks, 20 trials per entry. **Blue** frames: teleoperation sub-tasks; **red** frames: dexterous sub-tasks (kinesthetic).
- **Columns:** **Kin.** raw kinesthetic frames; **Mask-Kin.** kinesthetic + mask conditioning; **dash** sub-task taught by teleoperation.
- **No single modality** completes any task end-to-end.
- **Composition efficiency:** end-to-end success divided by the product of the sub-task success rates.

FUTURE WORK

- **Learn the decomposition:** sub-tasks and their modality, instead of specifying them by hand.
- **Learned handoff score:** a success predictor instead of the distance to start states.
- **Scale up:** more objects, longer sub-task chains, bimanual tasks.

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